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Batch: COMETFWC044
Date: 16 Feb 2026

Gate Question 2010 EE, Question No.52

Question

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

The following Karnaugh map represents a function F.

	YZ			
	00	01	11	10
$X = 0$	1	1	1	0
$X = 1$	0	0	1	0

Q.52 A minimized form of the function F is:

(A) $F = \overline{X}Y + YZ$

(B) $F = XY + YZ$

(C) $F = \overline{XY} + \overline{Y}Z$

(D) $F = X\overline{Y} + \overline{Y}Z$

Correct Answer: (B)

Question Analysis

From the K-map:

$$F = \sum m(0, 1, 3, 7)$$

Grouping gives:

$$F = XY + YZ$$

Truth Table

X	Y	Z	F
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Hardware Implementation

The minimized expression is implemented using Arduino UNO and a 7-segment display. Inputs X, Y, Z are manually connected. Output F is displayed as 0 or 1.

Required Components

- Arduino UNO
- Breadboard
- 7-Segment Display (Common Cathode)
- 220Ω Resistor
- Jumper Wires

Pin Connections

Inputs

Input	Arduino Pin
X	2
Y	3
Z	4

Open = 1
Connect to GND = 0

7-Segment Connections

Segment	Arduino Pin
a	5
b	6
c	7
d	8
e	9
f	10
g	11

Common Cathode → 220Ω → GND

Logic Description

$$F = XY + YZ$$

If $F = 1 \rightarrow$ Display 1
If $F = 0 \rightarrow$ Display 0

Code Uploading Steps

1. Create a Platform IO project.
2. Write the code in main.cpp inside the src folder.
3. Run the PIO project with command "pio run". It will compile the code and create a .hex file.
4. Copy the .hex file to the ArduinoDroid folder.
5. Connect the Arduino UNO to mobile using OTG cable.
6. Upload the .hex file using "Upload Precompiled" option.
7. Observe the output and verify the expression.

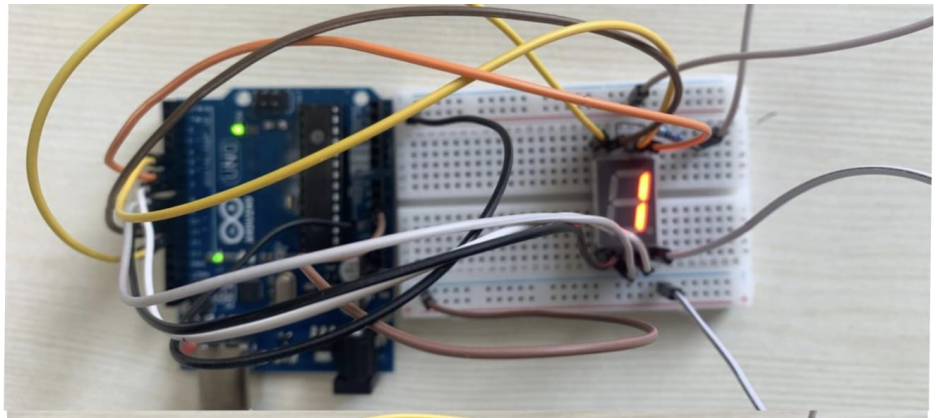
Experimental Truth Table

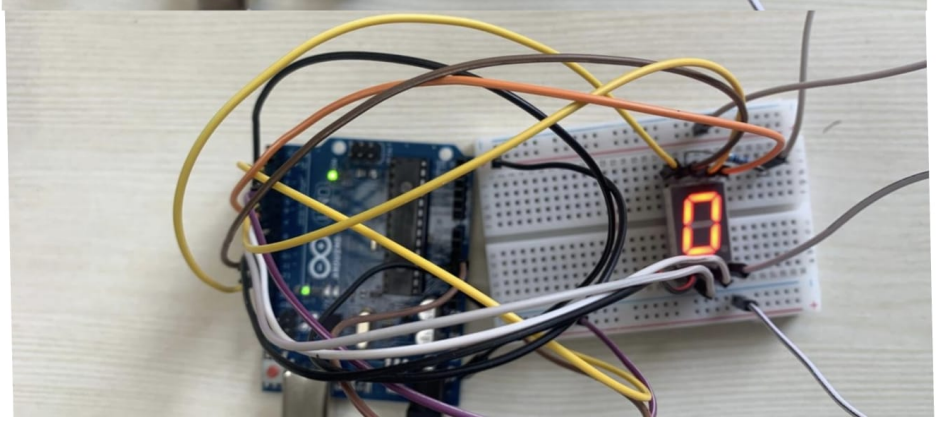
The experimentally verified truth table is shown below:

X	Y	Z	F (Observed)
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

The experimental results match the theoretical expression:

$$F = XY + YZ$$





Conclusion

The minimized Boolean expression of the given function is:

$$F = XY + YZ$$

This matches Option (B). Experimental output matches theoretical results.